

ArtiSG: Functional 3D Scene Graph Construction via Human-demonstrated Articulated Objects Manipulation

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Abstract—Functional 3D scene graphs localize actionable elements such as handles and knobs, but they describe only *where* these elements are, not *how* they move. We present ArtiSG, which augments functional 3D scene graphs with articulation recovered from human demonstrations. A demonstration serves as two complementary signals at once. It yields the kinematics of an element, sidestepping the visual ambiguity that defeats appearance-based inference, and its contact location acts as a detection prior that recovers inconspicuous elements, such as recessed handles, which open-vocabulary detectors miss entirely. A portable head-mounted camera tracking a rigid gripper interface records 6-DoF trajectories under free-space operator ego-motion, and both the fitted joint axes and the trajectories themselves are attached to element nodes. On real indoor scenes, contact-based refinement raises functional element recall from 55.8% to 88.5%, and the stored trajectories enable a robot to open articulated objects whose mechanisms are ambiguous from appearance alone.

I. INTRODUCTION

3D scene graphs abstract an environment into a compact, queryable map, and have supported language-guided retrieval [1], [2], navigation [3], [4], and planning [5], [6]. A recent line of work [7]–[9] pushes this representation below the object level, detecting the *functional elements* – handles, knobs, buttons – through which an agent actually interacts with the scene. However, these maps remain purely static. They record *where* an element is, but not *how* it moves, leaving a robot unable to tell a pull-out drawer from a flip-down door.

Recovering the missing kinematic layer from vision alone is hard. Appearance-based methods [10]–[12] predict motion from static observations, but near-identical appearances routinely hide distinct mechanisms [13]. Methods that observe a state change [14]–[16] are more accurate, yet they assume fixed viewpoints and unobstructed manipulation views, which are impractical in real, cluttered scenes. A second failure sits upstream of kinematics, where elements without distinctive visual signatures – a recessed grip, a push-to-open panel – can never be detected at all, so no motion reasoning can recover them.

Our core insight is that a single human demonstration can simultaneously address both challenges. The trajectory of the demonstrator’s hand naturally reveals the object’s kinematics, while the physical contact point serves as a powerful spatial prior that validates the presence of an otherwise inconspicuous functional element, regardless of its visual salience. Based on this insight, as shown in Fig. 1, we present ArtiSG, a framework that encodes observed human manipulations into a structured, semantic 3D scene graph. Serving as a robotic functional memory, the resulting graph directly guides downstream physical interactions with articulated objects.

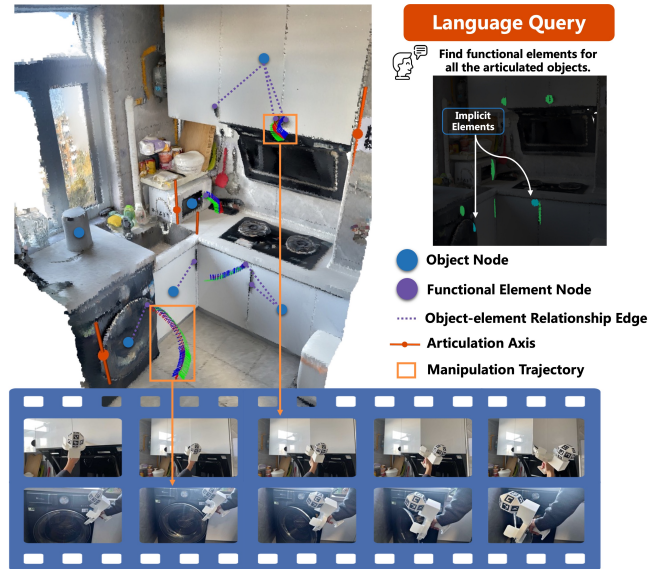


Fig. 1. **Augmenting functional scene graphs via human demonstration.** The bottom film strips show our manipulation sequences using a custom UMI gripper. From these sequences, we extract articulation trajectories and estimate axes, registering them to the corresponding element nodes in the graph. This structured representation enables open-vocabulary queries to locate functional elements and provides actionable priors for robot manipulation.

In summary, our contributions are as follows:

- We augment open-vocabulary functional 3D scene graphs with **articulation features**, linking both fitted joint models and raw 6-DoF trajectories directly to element nodes. This enables the graph to store immediately replayable manipulation memory rather than abstract kinematic parameters alone.
- We demonstrate that **physical contact acts as a powerful detection prior**, rather than merely a kinematics reference. By instantiating functional element nodes directly at the demonstrated contact coordinates, we increase real-world recall from 55.8% to 88.5%, successfully recovering inconspicuous elements missed by open-vocabulary detectors.
- We realize this with a **viewpoint-robust capture setup** that remains robust under free operator movement in complex real-world scenes, and deploy the resulting articulated scene graph on a physical robot for language-guided manipulation.

II. RELATED WORK

3D Scene Graphs. 3D scene graphs [1]–[6], [17] have facilitated task planning and navigation, though they typically

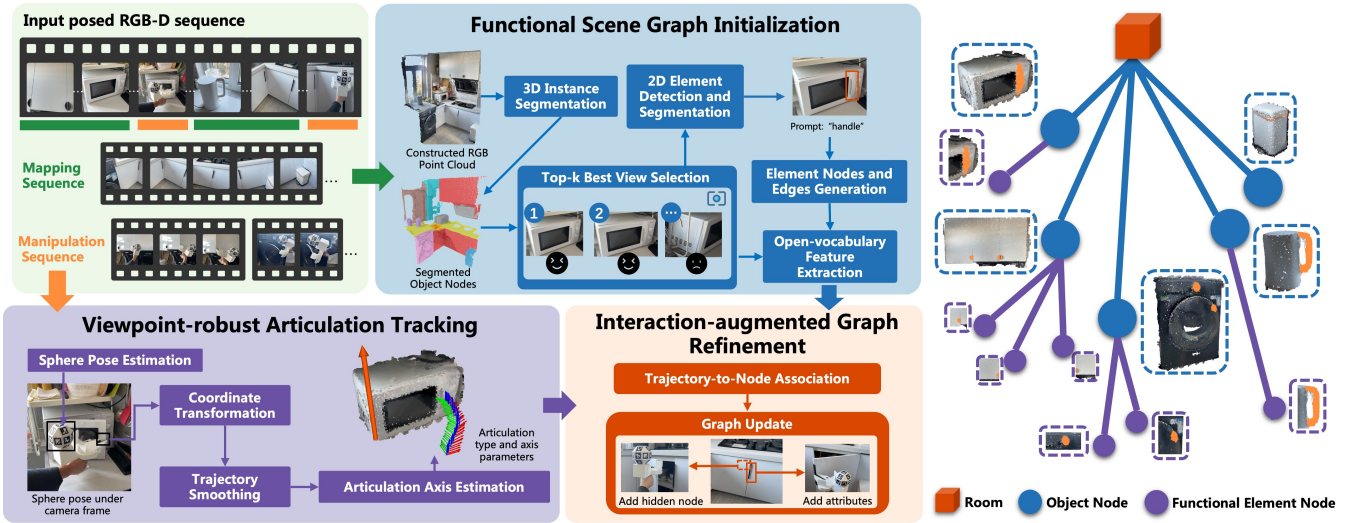


Fig. 2. **Framework overview.** ArtiSG constructs a functional 3D scene graph in three stages. First, we initialize object and element nodes by aggregating multi-view visual observations. Second, we estimate articulation parameters and 6-DoF trajectories by tracking human demonstrations with a portable interface. Finally, we refine the graph by instantiating missed, inconspicuous elements and attaching kinematic attributes directly to the element nodes.

focus on coarser-grained objects. To bridge this gap, recent methods [7]–[9] detect element-level components like handles and knobs, primarily focusing on static visual observations. Concurrently, MoMa-SG [18] and Pandora [19] incorporate articulation into scene graphs by estimating part-level joint axes and opening states from videos. ArtiSG builds upon these insights by constructing a hierarchical object-element graph, utilizing human contact locations as a prior to recover elements missed by semantic detectors and storing trajectories to enable direct robotic replay.

Articulated Object Understanding. Static methods [10]–[12], [20] predict contact poses and trajectories from appearance alone, which can be challenging when visually similar objects have different internal mechanisms. State-change methods [16], [21], [22] estimate joint parameters from pre-interaction and post-interaction states, often relying on fixed viewpoints or unobstructed observations. Concurrently, ArtiPoint [23] relaxes viewpoint constraints by estimating articulation from videos, though tracking textureless or occluded parts during dynamic interaction remains a challenge. ArtiSG addresses this by tracking a handheld gripper, reducing the reliance on continuous visual tracking of moving object parts.

Human-demonstrated Manipulation. While in-the-wild videos [12], [20], [24] scale but lack precision, and teleoperation [25] is precise but not portable, systems like UMI [26] and FastUMI [27] strike a balance with wearable hardware. However, they primarily rely on wrist-mounted cameras for visual policy learning. We instead design a decoupled setup optimized for functional scene understanding. By using a head-mounted SLAM camera to construct the static scene graph while tracking a handheld gripper, ArtiSG registers both static geometry and dynamic articulation in a unified coordinate frame.

III. APPROACH

We construct a functional 3D scene graph, formally defined as a tuple $\mathcal{G} = \{\mathcal{N}^{\text{obj}}, \mathcal{N}^{\text{ele}}, \mathcal{E}\}$, to represent indoor environments containing articulated objects. Each object node $N_i^{\text{obj}} \in \mathcal{N}^{\text{obj}}$ carries a category label, an open-vocabulary feature, and a point cloud. Each functional element node $N_j^{\text{ele}} \in \mathcal{N}^{\text{ele}}$ represents an actionable component (such as a handle or knob) and stores a functional label, an open-vocabulary feature, an articulation type, an articulation axis $\mathbf{A}_j = \{\mathbf{p}_c, \mathbf{p}_d\}$ with center $\mathbf{p}_c \in \mathbb{R}^3$ and direction $\mathbf{p}_d \in \mathbb{R}^3$, and a demonstrated 6-DoF trajectory $\mathcal{T}_j = \{\mathbf{p}_1, \dots, \mathbf{p}_n\}$ where each $\mathbf{p}_k \in \mathbb{R}^7$. Hierarchical edges $e_{ij} \in \mathcal{E}$ link elements to their parent objects in a one-to-many mapping. Fig. 2 illustrates the framework overview.

A. Functional Scene Graph Initialization

Object Node Construction. We process posed RGB-D sequences to construct a 3D point cloud of the scene. To extract object instances, we apply a 3D instance segmentation model [28] followed by DBSCAN clustering [29] to remove outliers, instantiating each denoised cluster as an object node N_i^{obj} in the graph.

Top- k Frame Selection. Since 2D models struggle with heavy occlusions on articulated parts, we select the most informative viewpoints for each object node N_i^{obj} . We project the 3D points of N_i^{obj} onto the 2D image plane of each camera frame t , removing points that fall outside image boundaries or exhibit occlusion-induced depth inconsistencies. We calculate a contribution score $s_{t,i}$ as the percentage of remaining valid projected points relative to the total object points, selecting the top- k frames with the highest scores for subsequent processing.

Element Node and Edge Construction. Within each cropped top- k frame of an object, we detect functional elements using Grounding DINO [30] using the prompts “handle”,

“knob” and “button”, and segment them with SAM [31]. These 2D masks are back-projected and aggregated in 3D to construct element nodes N_j^{ele} , a multi-view lifting strategy that successfully captures small functional details. Because elements are detected directly within the visual boundaries of their parent object N_i^{obj} , hierarchical edges e_{ij} are established naturally without a separate association step.

Open-vocabulary Feature Extraction. We extract open-vocabulary semantic features from the cropped top- k frames using SigLIP 2 [32]. These features are aggregated into a single node descriptor for both object and element nodes using a weighted average based on the contribution scores $s_{t,i}$. This ensures that viewpoints with higher visibility contribute more to the final representation, improving robustness against visual ambiguity.

B. Viewpoint-robust Articulation Estimation

Hardware Setup. Our framework utilizes a head-mounted RGB-D camera to track a handheld, 3D-printed parallel-jaw UMI gripper [26] equipped with a custom polyhedral sphere featuring ArUco markers (shown in the lower-left corner of Fig. 2). Tracking this rigid gripper instead of human hands bypasses complex hand-object contact variations, providing a consistent reference frame. The head-mounted camera uses built-in SLAM to track poses in a globally consistent world frame, ensuring robust 6-DoF trajectory recording under free operator ego-motion.

Trajectory Tracking. To recover the 6-DoF trajectory of the gripper tip in the world frame, we detect visible ArUco markers and establish 2D-3D correspondences between detected IDs $\{u_i^{2D}\}$ and pre-calibrated 3D corner positions $\{P_i^{3D}\}$. An initial local pose $T_{\text{cam} \leftarrow \text{sphere}} \in SE(3)$ is obtained via a Perspective-n-Point (PnP) solver [33]

$$T_{\text{cam} \leftarrow \text{sphere}} = \text{solvePnP}(\{P_i^{3D}\}, \{u_i^{2D}\}, K) \quad (1)$$

where K is the camera intrinsic matrix. We transform this local pose into the world frame using the SLAM camera pose $T_{\text{world} \leftarrow \text{cam}}$ via $T_{\text{world} \leftarrow \text{sphere}} = T_{\text{world} \leftarrow \text{cam}} \cdot T_{\text{cam} \leftarrow \text{sphere}}$. To suppress jitter, we apply an adaptive Kalman filter [34] with rotation unwrapping to ensure smooth transitions, weighting the filter confidence adaptively by the PnP reprojection error. Finally, we apply a static offset transformation $T_{\text{sphere} \leftarrow \text{tip}}$ to compute the gripper tip pose $T_{\text{world} \leftarrow \text{tip}} = T_{\text{world} \leftarrow \text{sphere}} \cdot T_{\text{sphere} \leftarrow \text{tip}}$, forming the final trajectory $\mathcal{T}_j$.

Articulation Axis Estimation. We estimate the kinematic mechanism as either a prismatic or revolute joint using Principal Component Analysis (PCA) and non-linear optimization. For a prismatic joint, we apply SVD to the centered points of $\mathcal{T}_j$, setting the direction $\mathbf{p}_d$ as the primary eigenvector and the center $\mathbf{p}_c$ as the trajectory centroid. For a revolute joint, the axis $\mathbf{p}_d$ is the normal to the fitted motion plane, and the rotation center $\mathbf{p}_c$ is solved on this plane using non-linear least squares. We select the final joint type by comparing reconstruction residuals, penalizing model complexity using the Bayesian Information Criterion (BIC) to prevent overfitting. This yields the articulation type and parameters $\mathbf{A}_j = \{\mathbf{p}_c, \mathbf{p}_d\}$.

TABLE I
FUNCTIONAL 3D SCENE GRAPHS EVALUATION

Scene	Method	Fun. Ele. Node			Overall Node	
		R _{>0}	P _{>0}	F1 _{>0}	R@1	R@5
Real-world	OpenFunGraph [8]	45.7	18.4	26.3	50.0	63.1
	FunGraph [7]	51.9	23.5	32.4	47.6	64.3
	ArtiSG (w.o top- k)	6.2	5.6	5.9	19.0	21.4
	ArtiSG (w.o human)	55.8	41.0	47.2	60.7	70.2
	ArtiSG (ours)	88.5	51.6	65.2	79.8	89.3
Simulation	OpenFunGraph [8]	39.7	66.1	49.6	34.0	41.8
	FunGraph [7]	56.1	74.3	63.9	44.4	49.7
	ArtiSG (w.o top- k)	34.7	64.8	45.2	28.0	35.3
	ArtiSG (w.o human)	78.6	70.4	74.2	59.0	73.2

C. Interaction-augmented Graph Refinement

Trajectory-to-Node Association. To unify coordinate systems between mapping and manipulation sequences, the camera scans a fixed environment ArUco marker to establish a global anchor frame. We align the trajectories and ground the dynamic data into the static graph by computing the Euclidean distance between the initial contact point $\mathbf{p}_1$ and the centroids of adjacent functional nodes. This distance is compared against a spatial threshold to identify matches or detect missed elements.

Graph Update. If the nearest node is within the spatial threshold, we attach the inferred axis $\mathbf{A}_j$, joint type, and trajectory $\mathcal{T}_j$ as dynamic attributes to that node. Otherwise, indicating the element has been missed during initialization due to occlusion or subtle appearance, we instantiate a new functional element node at the contact point $\mathbf{p}_1$ and link it to the nearest parent object node. This interaction-augmented update directly compensates for visual perception failures.

IV. EXPERIMENTS

We evaluate ArtiSG across scene graph construction quality, articulation tracking precision, and real-world robotic manipulation utility.

A. Functional Scene Graph Construction Evaluation

Setup. We evaluate ArtiSG on 79 articulated objects and 139 functional elements across real-world settings and three Behavior-1k [35] simulation scenes. We compare against OpenFunGraph [8] and FunGraph [7] and assess two ablations, with ArtiSG (w.o top- k) using random frames instead of viewpoint optimization, and ArtiSG (w.o human) relying purely on static visual scanning. Performance is measured using Precision (P), Recall (R), and F1-score for element detection under an IoU threshold of > 0 , alongside the node retrieval query success rate R@n.

Results. As shown in Table I, ArtiSG outperforms existing methods in both functional element detection and open-vocabulary node retrieval. Baselines rely solely on static appearance, struggling to detect implicit or recessed elements and limiting their real-world recall to 45.7% and 51.9%. In contrast, our static ablation ArtiSG (w.o human) achieves comparable performance to FunGraph by leveraging vision foundation models. Incorporating human interaction cues in

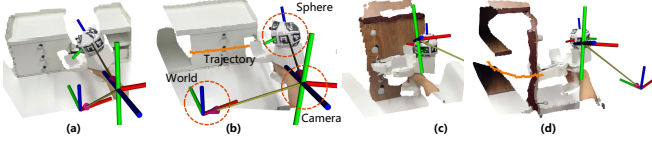


Fig. 3. **Visualization of the viewpoint-robust articulation tracking process.** Subfigures (a) and (b) depict the start and end phases of manipulating a prismatic joint, while (c) and (d) show the manipulation of a revolute joint. The distinct coordinate frames for the World, Camera, and Sphere are highlighted to illustrate our decoupled tracking setup. The recovered gripper trajectory is visualized as an orange curve, demonstrating smooth and precise tracking performance.

ArtiSG (ours) further increases real-world recall from 55.8% to 88.5%, proving that physical contact successfully uncovers visually hidden elements. For open-vocabulary node retrieval, ArtiSG consistently outperforms baselines in R@1 and R@5. This is driven by our top- k viewpoint selection, without which (as shown by the ArtiSG (w.o top- k) ablation) feature aggregation suffers from heavy noise and occlusions. This degradation is especially severe in real-world scenes where random viewpoint selection frequently misses key elements entirely.

B. Articulation Tracking Evaluation

Setup. For tracking evaluation, we utilize an iPhone 12 Pro for camera pose tracking and a UMI gripper with an ArUco sphere for end-effector tracking, obtaining ground-truth trajectories from an OptiTrack system. We test under static views where the operator is stationary and dynamic views where the operator moves naturally. We compare ArtiSG against GFlow [12], CoTracker [36], ArtiPoint [23], and Mediapipe [37]. To ensure a fair comparison, since the gripper occludes the hand and target surface, we evaluate baselines on bare-hand sequences and pass their trajectories to our axis estimation module. We report the trajectory RMSE T_{err} , axis angular error θ_{err} , and axis position error d_{err} .

Results. As shown in Fig. 3 and Table II, ArtiSG achieves high accuracy across static and dynamic settings. Under static views, GFlow exhibits high angular error, highlighting the ambiguity of predicting kinematics from static visual observations. Compared to hardware-free baselines, our rigid-body tracking yields significantly lower errors. While concurrent work ArtiPoint relaxes tracking hardware constraints, it exhibits high angular and position errors. Compared to CoTracker and Mediapipe, ArtiSG reduces the static revolute trajectory RMSE from 7.31 cm and 3.83 cm to 1.09 cm, demonstrating that tracking a rigid gripper bypasses hand-tracking jitter and point-tracking occlusions. ArtiSG also remains robust under free operator movement due to decoupled SLAM-based tracking. Finally, while most baselines perform acceptably on simpler prismatic joints, ArtiSG excels on complex revolute joints requiring precise arc fitting. In dynamic revolute joint manipulation, our method reduces the trajectory RMSE by approximately 70% compared to Mediapipe, confirming its suitability for real-world articulation tracking.

TABLE II
QUANTITATIVE COMPARISON OF ARTICULATION ESTIMATION PERFORMANCE.

Setting	Method	Prismatic joints		Revolute joints		
		T_{err} (cm) ↓	θ_{err} (deg) ↓	T_{err} (cm) ↓	θ_{err} (deg) ↓	d_{err} (cm) ↓
Static	GFlow [12]	-	16.610	-	38.084	13.122
	ArtiPoint [23]	-	16.334	-	11.731	22.700
	CoTracker [36]	14.342	4.145	7.310	4.976	1.883
	Mediapipe [37]	1.788	3.703	3.826	4.066	2.219
	ArtiSG(ours)	0.976	1.026	1.092	1.627	0.811
Dynamic	ArtiPoint [23]	-	17.361	-	39.326	19.281
	CoTracker [36]	7.967	1.541	11.039	5.016	3.619
	Mediapipe [37]	1.083	2.291	2.953	6.644	3.757
	ArtiSG(ours)	0.820	1.314	0.899	2.322	1.225

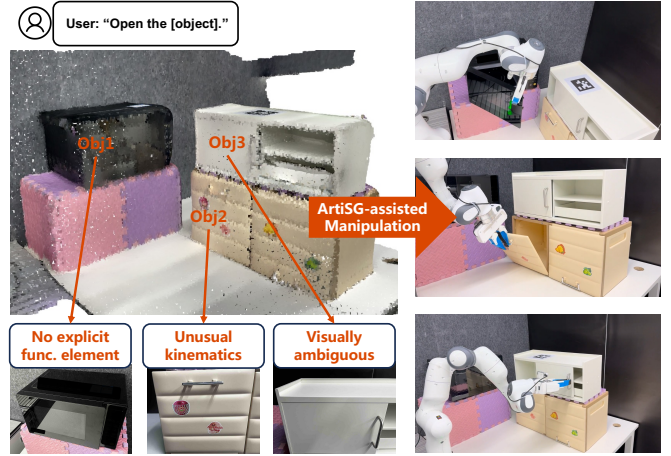


Fig. 4. **ArtiSG-assisted manipulation.** Objects chosen for inconspicuous elements (Obj1), unusual kinematics (Obj2) and visual ambiguity (Obj3). Inference from static observation misreads these mechanisms, treating the flip-down door of Obj2 as a drawer, whereas retrieving the stored demonstration yields a 6-DoF trajectory that opens each object.

C. ArtiSG-assisted Robot Manipulation Demonstration

We evaluate the physical utility of ArtiSG using a Franka Research 3 robot tasked with opening three challenging articulated objects that defeat static visual perception (Fig. 4). The robot resolves the natural language query against the node features of our scene graph, retrieves the stored 6-DoF trajectory, and replays it. Since the joint kinematics are directly observed from human demonstrations rather than predicted from static images, the visual ambiguity that typically defeats appearance-based reasoning is bypassed, allowing the robot to execute the correct physical motions.

V. CONCLUSION

In this work, we presented ArtiSG, a framework that augments functional 3D scene graphs with articulation trajectories recovered from human demonstrations to bypass static visual ambiguity. In the future, we plan to develop a lighter, markerless tracking interface and integrate our articulated scene graphs with vision-language-action (VLA) models as affordance guidance for complex robotic manipulation.

REFERENCES

- [1] L. Ge, X. Zhu, Z. Yang, and X. Li, "DynamicGSG: Dynamic 3d gaussian scene graphs for environment adaptation," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 2232–2239.
- [2] Q. Gu, Z. Ye, J. Yu, J. Tang, T. Yi, Y. Dong, J. Wang, J. Cui, X. Chen, and Y. Wang, "MR-COGraphs: Communication-efficient multi-robot open-vocabulary mapping system via 3d scene graphs," *IEEE Robotics and Automation Letters*, 2025.
- [3] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard, "Hierarchical open-vocabulary 3d scene graphs for language-grounded robot navigation," *Robotics: Science and Systems*, 2024.
- [4] J. Chen, G. Li, S. Kumar, B. Ghanem, and F. Yu, "How to not train your dragon: Training-free embodied object goal navigation with semantic frontiers," *Proceedings of Robotics: Science and System XIX*, p. 075, 2023.
- [5] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa *et al.*, "Conceptgraphs: Open-vocabulary 3d scene graphs for perception and planning," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 5021–5028.
- [6] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Suen-derhauf, "Sayplan: Grounding large language models using 3d scene graphs for scalable robot task planning," in *7th Annual Conference on Robot Learning*, 2023.
- [7] D. Rotondi, F. Scaparro, H. Blum, and K. O. Arras, "Fungraph: Functionality aware 3d scene graphs for language-prompted scene interaction," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- [8] C. Zhang, A. Delitzas, F. Wang, R. Zhang, X. Ji, M. Pollefeys, and F. Engelmann, "Open-Vocabulary Functional 3D Scene Graphs for Real-World Indoor Spaces," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [9] J. Corsetti, F. Giuliani, A. Fasoli, D. Boscaini, and F. Poiesi, "Functionality understanding and segmentation in 3d scenes," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- [10] R. Wu, Y. Zhao, K. Mo, Z. Guo, Y. Wang, T. Wu, Q. Fan, X. Chen, L. Guibas, and H. Dong, "VAT-mart: Learning visual action trajectory proposals for manipulating 3d articulated objects," in *International Conference on Learning Representations*, 2022. [Online]. Available: <https://openreview.net/forum?id=iEx3PiooLy>
- [11] X. Zhang, Y. Wang, R. Wu, K. Xu, Y. Li, L. Xiang, H. Dong, and Z. He, "Adaptive articulated object manipulation on the fly with foundation model reasoning and part grounding," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 13 032–13 042.
- [12] C. Yuan, C. Wen, T. Zhang, and Y. Gao, "General flow as foundation affordance for scalable robot learning," in *Conference on Robot Learning*. PMLR, 2025, pp. 1541–1566.
- [13] Y. Li, W. H. Leng, Y. Fang, B. Eisner, and D. Held, "Flowbothd: History-aware diffuser handling ambiguities in articulated objects manipulation," in *Conference on Robot Learning*. PMLR, 2025, pp. 5271–5293.
- [14] Z. Jiang, C.-C. Hsu, and Y. Zhu, "Ditto: Building digital twins of articulated objects from interaction," in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [15] J. Liu, A. Mahdavi-Amiri, and M. Savva, "PARIS: Part-level reconstruction and motion analysis for articulated objects," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2023.
- [16] Y. Liu, B. Jia, R. Lu, J. Ni, S.-C. Zhu, and S. Huang, "Building interactable replicas of complex articulated objects via gaussian splatting," in *The Thirtieth International Conference on Learning Representations*, 2025.
- [17] I. Armeni, Z.-Y. He, J. Gwak, A. R. Zamir, M. Fischer, J. Malik, and S. Savarese, "3d scene graph: A structure for unified semantics, 3d space, and camera," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 5664–5673.
- [18] M. Büchner, A. Röfer, T. Engelbracht, T. Welschhold, Z. Bauer, H. Blum, M. Pollefeys, and A. Valada, "Articulated 3D scene graphs for open-world mobile manipulation," *arXiv preprint arXiv:2602.16356*, 2026.
- [19] A. Yu, Y. Chang, C. Xie, and L. Carlone, "Pandora: Articulated 3D scene graphs from egocentric vision," *arXiv preprint arXiv:2603.28732*, 2026.
- [20] Y. Kuang, J. Ye, H. Geng, J. Mao, C. Deng, L. Guibas, H. Wang, and Y. Wang, "RAM: Retrieval-based affordance transfer for generalizable zero-shot robotic manipulation," in *Conference on Robot Learning*. PMLR, 2025, pp. 547–565.
- [21] C.-C. Hsu, Z. Jiang, and Y. Zhu, "Ditto in the house: Building articulation models of indoor scenes through interactive perception," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [22] C. Zhang and G. H. Lee, "Iaao: Interactive affordance learning for articulated objects in 3d environments," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 12 132–12 142.
- [23] A. Werby, M. Buechner, A. Roefer, C. Huang, W. Burgard, and A. Valada, "Articulated object estimation in the wild," *Conference on Robot Learning (CoRL)*, 2025.
- [24] Y. Ju, K. Hu, G. Zhang, G. Zhang, M. Jiang, and H. Xu, "Robo-abc: Affordance generalization beyond categories via semantic correspondence for robot manipulation," in *European Conference on Computer Vision*. Springer, 2025, pp. 222–239.
- [25] S. Wu, Y. Zhu, Y. Huang, K. Zhu, J. Gu, J. Yu, Y. Shi, and J. Wang, "Afforddp: Generalizable diffusion policy with transferable affordance," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 6971–6980.
- [26] C. Chi, Z. Xu, C. Pan, E. Cousineau, B. Burchfiel, S. Feng, R. Tedrake, and S. Song, "Universal manipulation interface: In-the-wild robot teaching without in-the-wild robots," in *Proceedings of Robotics: Science and Systems (RSS)*, 2024.
- [27] Zhaxizhuoma, K. Liu, C. Guan, Z. Jia, Z. Wu, X. Liu, T. Wang, S. Liang, P. CHEN, P. Zhang, H. Song, D. Qu, D. Wang, Z. Wang, N. Cao, Y. Ding, B. Zhao, and X. Li, "FastUMI: A scalable and hardware-independent universal manipulation interface with dataset," in *9th Annual Conference on Robot Learning*, 2025. [Online]. Available: <https://openreview.net/forum?id=RUSscFSEfD>
- [28] J. Schult, F. Engelmann, A. Hermans, O. Litany, S. Tang, and B. Leibe, "Mask3D: Mask Transformer for 3D Semantic Instance Segmentation," 2023.
- [29] M. Ester, H.-P. Kriegel, J. Sander, X. Xu *et al.*, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *kdd*, vol. 96, no. 34, 1996, pp. 226–231.
- [30] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, Q. Jiang, C. Li, J. Yang, H. Su, J. Zhu, and L. Zhang, "Grounding DINO: Marrying DINO with Grounded Pre-training for Open-Set Object Detection," in *Computer Vision – ECCV 2024*, A. Leonardis, E. Ricci, S. Roth, O. Russakovsky, T. Sattler, and G. Varol, Eds. Cham: Springer Nature Switzerland, 2025, pp. 38–55.
- [31] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo *et al.*, "Segment anything," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 4015–4026.
- [32] M. Tschannen, A. Gritsenko, X. Wang, M. F. Naeem, I. Alabdulmohsin, N. Parthasarathy, T. Evans, L. Beyer, Y. Xia, B. Mustafa, O. Hénaff, J. Harmsen, A. Steiner, and X. Zhai, "Siglip 2: Multilingual vision-language encoders with improved semantic understanding, localization, and dense features," *arXiv preprint*, 2025, arXiv:2502.14786 [cs.CV].
- [33] G. Bradski, "The OpenCV library," *Dr. Dobbs's Journal of Software Tools*, vol. 25, no. 11, pp. 120–125, 2000.
- [34] R. E. Kalman, "A new approach to linear filtering and prediction problems," 1960.
- [35] C. Li, R. Zhang, J. Wong, C. Gokmen, S. Srivastava, R. Martín-Martín, C. Wang, G. Levine, W. Ai, B. Martinez, H. Yin, M. Lingelbach, M. Hwang, A. Hiranaka, S. Garlanka, A. Aydin, S. Lee, J. Sun, M. Anvari, M. Sharma, D. Bansal, S. Hunter, K.-Y. Kim, A. Lou, C. R. Matthews, I. Villa-Renteria, J. H. Tang, C. Tang, F. Xia, Y. Li, S. Savarese, H. Gweon, C. K. Liu, J. Wu, and L. Fei-Fei, "Behavior-1k: A human-centered, embodied ai benchmark with 1,000 everyday activities and realistic simulation," *arXiv preprint arXiv:2403.09227*, 2024.
- [36] N. Karaev, I. Makarov, J. Wang, N. Neverova, A. Vedaldi, and C. Rupprecht, "CoTracker3: Simpler and better point tracking by pseudo-labelling real videos," in *Proc. arXiv:2410.11831*, 2024.
- [37] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee *et al.*, "Mediapipe: A framework for building perception pipelines," *arXiv preprint arXiv:1906.08172*, 2019.